

SVF (Cine Distribution) — Project & Development Report

End-to-end account of the platform: requirements, technology, and everything built from day one to now.

Coverage: 16 July 2026 (first commit) → 19 August 2026 · ~5 weeks · Solo build · Live on Vercel

What SVF is. SVF is an internal web platform for a film-distribution operation covering the Nizam theatre circuit. It brings three products into one codebase: an **Admin console** (theatres, representatives, roles, rate cards, users), a **Representative app** for theatre reps to submit daily sales reports, and a **Movies workspace** that turns an uploaded Excel daily-collections report into a live, Excel-faithful, shareable spreadsheet. It is now **live in production** and being prepared for real-world use during a movie release.

1. Requirements & Objectives

The distribution operation ran on manual processes and standalone Excel files. The goal was a single, secure web platform that digitises the whole workflow — from the theatre master data down to the daily collection reports — while keeping the familiar Excel look-and-feel the team already relies on.

- Functional requirements**
- Master catalogue of every theatre in the Nizam circuit (district → centre → theatre → screen) with seating capacity and full-house values.
 - Manage field representatives and assign them to theatres.
 - Role- and permission-based access for admin staff.
 - Per-screen rate cards (rate slabs, full-house values).
 - A guided, mobile-friendly flow for reps to submit daily show/sales reports from the field.
 - Open the business's formatted daily-collections **Excel files online** — view and edit them exactly as they look in Excel.
 - Share those sheets with the right people and collaborate in real time.
 - Export reports back to **Excel** and **PDF**.
 - Support multiple user types (admin, distribution users, representatives).

- Non-functional requirements**
- **Fidelity** — the online sheet must match the source Excel precisely (colours, borders, merged cells, number formats).
 - **Security** — proper authentication and role-based access across all three user types.
 - **Reliability** — dependable auto-save and safe multi-user collaboration.
 - **Performance** — fast to open and responsive to use.
 - **Access control** — viewers strictly read-only; editors can change; owners/admin manage sharing.
 - **Deployability** — continuously deployed and stable in production for use during live movie releases.

2. Technology Stack

LAYER	TECHNOLOGY	WHY / WHAT IT DOES
Framework	Next.js 15 (App Router)	Full-stack React framework — pages, routing, and API endpoints in one app
UI	React 19, TypeScript 5, Tailwind CSS 3.4	Component UI, type safety, and styling

LAYER	TECHNOLOGY	WHY / WHAT IT DOES
Spreadsheet engine	Univer (@univerjs/presets)	Canvas-based spreadsheet that renders/edits the uploaded Excel in the browser
Excel input	ExcelJS, xlsx	Parse uploaded .xlsx files (values, styles, theme, merges)
Export	jsPDF + AutoTable, ExcelJS	Download sheets back to PDF and Excel
Database	PostgreSQL (Neon), via pg (SSL, pooled)	Shared data store for theatres, movies, sheets, shares, reps, reports
Authentication	Custom HMAC-SHA256 sessions; scrypt hashing (reps)	Secure sign-in for admin, users, and reps — no third-party auth library
Rep app routing	react-router-dom	Self-contained SPA for the representative flow
Hosting / CI-CD	Vercel (GitHub-linked)	Automatic build & deploy on every push to main

3. Architecture at a Glance

Three products, one codebase, one database. A single Next.js app serves the Admin console, the Representative SPA, and the Movies workspace, all backed by one PostgreSQL database. Access is separated by **three independent signed sessions** — admin, user, and rep — plus an admin "Login as" impersonation mode for support.

Core data model (PostgreSQL)

TABLE	PURPOSE	KEY COLUMNS
theatres	Nizam theatre catalogue (seeded from data file)	district, centre, theatre, format, screen_count, capacity, screens
movies	Report / movie metadata	id, name, release, owner_email
movie_sheets	The spreadsheet content	movie_id, data (JSON), version
movie_shares	Per-user access to a movie	(movie_id, email), role (editor/viewer)
reps	Field representatives	id, name, email, region, status, role_id
reports	Rep daily sales submissions	rep_id, theatre/screen/movie, seats & collection totals, occupancy

4. How the Work Progressed (at a glance)

PHASE	DATES	FOCUS
Week 1 — Foundations	16–22 Jul	Full-stack setup, admin console, representative app, first live deploy
Week 2 — Spreadsheet platform	22–24 Jul	Excel-to-spreadsheet workspace, user accounts, sharing, export
Week 3 — Rendering fidelity	27–29 Jul	Render uploaded Excel as an exact visual replica
Week 4 — Editing & access control	30 Jul–1 Aug	Excel-parity editing tools, viewer read-only, keyboard fixes
Week 5 — Hardening & go-live	1–19 Aug	Production fixes, sharing refinements, performance, launch

5. Detailed Timeline

Week 1 — Foundations & the V1 Platform

16–22 Jul 2026

Goal: stand up the whole platform skeleton and get the first two products working end-to-end.

- **Project foundation** — Next.js + React + TypeScript + Tailwind app; PostgreSQL connected; Vercel deployment pipeline.
- **Custom authentication** — signed sessions for admin, users, and reps, plus admin "Login as" impersonation (no third-party auth library).
- **Admin console** — admin login; the full Nizam **theatre catalogue** (district → centre → theatre → screens, with capacity) seeded into the database; dashboard/coverage; representatives management; roles & permissions; rate cards.
- **Representative app** — a mobile-style app for reps: login, then a guided flow (assigned theatres → show selection → sales entry → review → success), with automatic calculations and submission to the database.
- **First production deploy** — cleared build blockers so the app deployed cleanly to Vercel.

Week 2 — The Spreadsheet Workspace, Accounts & Sharing

22–24 Jul 2026

Goal: build the flagship feature — turn uploaded Excel reports into live, collaborative spreadsheets.

- **Live spreadsheet workspace** — upload an Excel (.xlsx) daily-collections report and open it as a fully interactive spreadsheet in the browser.
- **Excel → spreadsheet converter** — the engine that parses an uploaded workbook and rebuilds it faithfully in the app.
- **User accounts & access** — app-user login; admin-managed users; "Login as" to view the app exactly as any user.
- **Sharing & collaboration** — share a movie as editor or viewer; automatic saving; multi-user updates with clients polling for others' edits.
- **Export** — download a sheet back to Excel.
- **Stability** — fixed movie-open crashes, blank-collaboration states, and added loading indicators.

Week 3 — Excel Rendering Fidelity

27–29 Jul 2026

Goal: make an uploaded report look exactly like it does in Excel.

- **Exact visual replica** — preserved values, formulas, fill colours, fonts, borders, alignment, number formats, merged cells, column widths, row heights, and tab colours.
- **True colours** — read each file's own colour theme so fills and fonts match the source precisely.
- **Clean grid** — uniform gridlines and the grid trimmed to the real data area (no wall of empty columns).
- **Repeated refinement passes** — several rounds of converter and rendering fixes to close remaining visual gaps.

Week 4 — Editing Experience, Access Control & Polish

30 Jul – 1 Aug 2026

Goal: make the spreadsheet feel like Excel to work in, and lock down what each role can do.

- **Format Painter** — full Excel parity: click to copy a cell's formatting to the next selection, double-click to keep applying, Esc to stop. Copies all formatting including row height and column width, never the value; a single Undo reverts a whole paint.
- **Excel "Standard Colors"** — added the standard colour row to the fill/font colour pickers.
- **Keyboard shortcuts** — fixed delete-whole-word (Ctrl+Shift+Backspace) and Undo/Redo (Ctrl+Z).
- **Hidden sheets** — hidden sheets stay hidden on upload and can be revealed via right-click → Unhide, like Excel.
- **Viewer read-only mode** — viewers get a true view-only screen: no toolbars, no formula bar, no editing, no reset, formulas hidden.
- **Data-entry stability** — fixed a bug where the selected cell could jump elsewhere while entering data.
- **Documentation** — produced end-to-end project documentation.

Week 5 — Production Hardening, Sharing Refinements & Go-Live

1–19 Aug 2026

Goal: take the platform to production and make it reliable and fast for real use.

- **Went live in production** — ready for real-world testing during a movie release.
- **Production reliability fix** — the "Standard Colors" feature worked locally but not in production; re-engineered it to ship reliably with every deployment.
- **Sharing refinements** — the "Shared with you" list now attributes a sheet to the person who actually shared it (never the internal admin account), and hides the label when there is no real sharer; sharing was also unlinked from the admin page.
- **Performance** — sheets open noticeably faster by warming up the spreadsheet engine in the background.
- **Operational polish** — assorted clean-ups and dev/prod troubleshooting to keep the live app stable.

6. The Platform Today — Feature Summary

Admin console

- Theatre catalogue (Nizam circuit) with capacity data
- Representatives & assignments
- Roles & permissions
- Rate cards

Representative app

- Rep login (secure, DB-validated)
- Guided daily-report flow
- Automatic sales / occupancy calculations
- Report history

- User management + "Login as" impersonation

Movies workspace

- Upload Excel → exact live spreadsheet
- Full editing with Excel-parity tools
- Auto-save + real-time collaboration
- Editor / viewer sharing
- Export back to Excel & PDF

Platform & foundation

- Custom authentication (3 session types)
- PostgreSQL data layer
- Continuous deployment on Vercel
- End-to-end project documentation

7. Current Status & Next Steps

- **Status:** live in production; all three products (admin, rep app, movies workspace) complete and functional.
- **In final review:** a few recent refinements (sharing attribution, display clean-up, open-speed) are ready and pending the next deployment.
- **Next:** live testing with real daily-collections data during the upcoming release, then performance/bug monitoring and any field-driven improvements.